10/12 ch.2.5

DEFINITION Let A be an $m \times n$ matrix. Let $\mathbf{x}$ be an element of $\mathbf{R}^n$ written in column matrix form. A defines a *matrix transformation* $T(\mathbf{x}) = A\mathbf{x}$ of $\mathbf{R}^n$ into $\mathbf{R}^m$. The vector $A\mathbf{x}$ is the *image* of $\mathbf{x}$. The *domain* of the transformation is $\mathbf{R}^n$ and the *codomain* is $\mathbf{R}^m$.

Matrix transformations map line segments into line segments (or points). If the matrix is invertible the transformation also maps parallel lines into parallel lines.

* Not any transformation can be written as matrix.

Orthogonal matrix: $A^{-1} = A^t$

Orthogonal transformation. (rigid motion)

We leave it for the reader to show that orthogonal transformations also preserve angles and distances in Exercise 21.

Translation (平移): $T(u) = u + V \Rightarrow T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Affine transformation: A matrix transformation followed by a transition.

$$T(u) = Au + V$$

10/13
c.h.2.6

Linear transformation: ① $T(u+v) = T(u) + T(v)$ ② $cT(u) = T(cu)$

Standard matrix:

ch.3-1

Determinant: Sum of products of the elements of the first row and their cofactors.

2×2 matrix $A_{2\times 2} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow \det(A) = |A| = ad - bc$

Minor: M_{ij} is determinant of the matrix that remains after deleting row i and column j of A

e.g. $A = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 5 & 6 \\ 1 & 1 & 8 \end{bmatrix}$ $M_{21} = \begin{vmatrix} 3 & 4 \\ 1 & 8 \end{vmatrix} = \begin{vmatrix} 3 & 4 \\ 0 & 8 \end{vmatrix} = -4$

Cofactor: $C_{ij} = (-1)^{i+j} M_{ij}$ e.g. $C_{21} = (-1)^{2+1} \cdot (-4) = 4$